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Automation, artificial intelligence and capital concentration – A race for the machine*

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ABSTRACT

Since the industrial revolution, productive capital in the form of machines, structures and systems has assumed an ever more dominant role as compared to labour. The owners of non-human productive assets receive a growing share of market-sourced income, with an ever-smaller proportion going to the owners of human labour. The shift between factors of production in favour of capital is exacerbated by technological progress and its embodiment in automation. Academic work on labour's declining income share has emphasised the role of capital accumulation and capital-augmenting technical change exacerbated by the fast developments of artificial intelligence. At the same time, we observe a peculiar disconnect between this phenomenon and public discourse and interpretation of its social and economic effects:

- the discussion of automation is focused on its potential effects on employment, micro- and macro-distributive effects are barely mentioned.
- the discussion of rising inequality is focused on issues of distributive justice; the role of the shift from labour to capital in concentrating wealth is mostly ignored.

We argue that to offset the disadvantages of AI/automation and further capital concentration, all citizens should be enabled to (co-) own technology and benefit from the second income source capital ownership provides.

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1. Background

The issue of automation's distributive effects was already raised on both sides of the iron curtain in the aftermath of World War II. *Stanisław Lem*, the Polish writer and futurist, anticipated these problems in Poland as early as 1954 (Lem 1954). Shortly thereafter, in 1958, the American corporate lawyer and merchant banker *Louis O. Kelso* presented an analysis of and a proposal for dealing with this problem in his book 'The Capitalist Manifesto' (Kelso and Adler 1958). His solution to the problem was to broaden the

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ownership of productive capital through democratising access to capital credit, the key to acquiring productive property, in other words the non-human things that produce wealth in an industrial society. To accomplish this, Kelso invented the prototype of the leveraged buy-out and applied it to different constituencies – Employee Stock Ownership Plans (ESOPs) for corporate employees, Consumer Stock Ownership Plans (CSOPs) for consumers and General Stock Ownership Plans (GSOPs) for citizens in general.[1]

More than 60 years later, in 2014, the European Commission launched one of the world's largest civilian research and innovation programmes in robotics with the aim of Europe becoming *'a producer and not merely a consumer of robots'*. Kelso would have added his own postulate, urging for a 'race for the machine' to enable citizens to become owners of the robots. Against this background and in light of the decline of the wage share, this paper investigates fundamental questions of ownership, automation, income distribution and capital concentration. In an age of globalisation and technological progress, the acceleration of digital information and communication technologies (ICT) has led and continues to lead to social changes which shake the foundations of the world of work, dubbed 'industrial revolution 4.0'.[2] This process mainly characterised by automation, robotics and artificial intelligence impacts not only national law and economic systems but also traditional institutions of social existence and, therefore, the foundations of our modern societies in Europe and across the world. But unlike previous industrial revolutions, AI seems to have a different momentum, diffusing much faster than previous disruptive technologies.[3]

In its resolution of 16 February 2017 on Civil Law Rules on Robotics (2017), while focussing on issues of liability, cyber security and ethics, the European Parliament recommends that researchers in the field of robotics should commit to four core principles, one being *'Justice – fair distribution of the benefits associated with robotics'*. This notion of fairness and distributional justice goes back to a 2015 report of the Committee on Legal Affairs on Civil Law Rules on Robotics of the European Parliament (2015/2103(INL)) raising questions about *'the future of employment and the viability of social security systems'* and *'the potential for increased inequality in the distribution of wealth and influence'*. Consequently, the brand-new European Artificial Intelligence Act of 13 June 2024 (European Union 2024)[4] acknowledges in recital (142) possible distributional effects and calls for further research of these issues:

> 'To ***ensure that AI leads to socially and environmentally beneficial outcomes***, Member States are encouraged to support and promote ***research and development of AI solutions in support of socially and environmentally beneficial outcomes*** such as AI-based solutions to increase accessibility for persons with disabilities, **tackle socio-economic inequalities**, or meet environmental targets, by allocating sufficient resources, including public and Union funding, and, where appropriate and provided that the eligibility and selection criteria are fulfilled, considering in particular projects which pursue such objectives. Such projects should be based on the principle of interdisciplinary cooperation between AI developers, **experts on inequality and non-discrimination,** accessibility, consumer, environmental, and digital rights, as well as academics'. (our emphasis)

Property ownership is indeed the distributional mechanism of the free market which awards income from production to the producers according to their respective productive inputs: to landowners, the earnings of land; to workers, the earnings of labour; to capital owners, the earnings of capital. Over time productive capital – machines,

structures and systems – has assumed an ever more dominant role in the industrial production process. As a result, the owners of these productive assets – now including robots and AI – receive a growing share of market-sourced income, with an ever-smaller proportion going to the owners of human labour power.

Since money spent on goods and services is what keeps an industrial society running, the steady erosion of labour earnings is a problem that ultimately threatens the well-being of us all. But automation is already affecting our social security systems. According to official federal 2022 estimates in Germany, the average cost of an industrial working hour in 2022 was 40 Euros (Federal Statistical Office 2023) of which around a third is contributed to the social security system; the cost of a robot working hour comes to around 9 to 11 Euros,[5] including the costs of maintenance and repair. Proposals to compensate for the lost social security contributions by introducing a 'robot' or 'machine tax' have been made since the 1970s, most recently by (Carbonara, Focacci, and Santarelli 2023; Holzer 2021; Thuemmel 2023). Another debated idea is whether a Universal Basic Income (UBI) could compensate workers for losses of income, although critics object that a UBI would make workers even more dependent on the welfare state, while the concentration of income and wealth would continue perpetuating economic inequalities (Hall et al. 2019).[6]

Already in the wake of the 2008/09 financial crisis but in particular in the aftermath of the COVID-19 pandemic, new questions have arisen about automation's repercussions on the welfare state, e.g. why has median income stopped rising and why, as a consequence, have our economies and societies seemed to have gradually become more unequal? One widely discussed explanation is the thesis of French economist Thomas Piketty, namely that wealth becomes increasingly concentrated as the rate of return on capital is consistently higher than economic growth (Piketty and Goldhammer 2014). In short, as capital owners accumulate ever more productive assets, thereby increasing their capital income, workers receive an ever-shrinking share of labour income.

2. Five decades of decline of the labour share – where are we today?

The shift between factors of production in favour of capital is exacerbated by technological progress and its embodiment in automation. The 'race against the machine' is back in the headlines,[7] a competition perceived to threaten labour market equilibrium and – some pessimistic voices (Rifkin 2004)[8] fear – the future of labour employment. A literature review (Aghion et al. 2022, 24) on the topic showed that early studies were rather pessimistic, predicting a decline in employment even when considering the creation of new employment opportunities through automation. More recent studies, on the other hand, painted a more optimistic picture arguing that by increasing their productivity through automation, companies would gain market share and hire more staff; however, it should be noted that these studies also consider a possible decline in employment at the macro level, as the increase in personnel would then only apply to the automating companies, with non-automating companies being squeezed out of the market and reducing their workforce. In this light Aghion et al. (2022, 24) conclude that with the positive view primarily looking at the level of the individual company, empirical data for the macroeconomic level points to a decline in employment.

It appears that digital ICT have altered the very nature of technological change; the most recent advances are incorporated in machines and devices whose main purpose is to replace human participation entirely thereby eliminating labour jobs (Brynjolfsson and McAfee 2011, 2014; McAfee and Brynjolfsson 2018). Digital innovation is a phenomenon both broad and deep; its economic implications are profound. Many of these are positive, i.e. productivity increases, price reductions and a growing economic pie. There are important distributive effects, however, which are unfavourable – above all for those whose skills computers have mastered and who now see their wages and opportunities shrinking. According to estimates from a study by Oxford economists Frey and Osborne (2017) in the US labour market, 47% of employment could potentially be automated in the next two decades. These findings are corroborated by a World Economic Forum (2023), assessing 42% of all business tasks likely to be fully automated by 2027. A study by Acemoglu and Restrepo (2020), examining US data from 1990 to 2007, estimated that one more robot per 1,000 workers triggered a fall in population-to-employment ratio of 0.2% points; aggregated wages during this interval fell by 0.42% per robot with each robot replacing 3.3 workers. If this process were to continue unchecked, most workers, particularly the least educated who are unable to adapt quickly, would find themselves in a competition with machines, their relative positions ever worse.

Indeed, recent academic studies on the decline of labour's income share (see Figure 1) point to the role of capital accumulation and capital-augmenting technical change (see, e.g. Arpaia, Pérez, and Pichelmann 2009; Bentolila and Saint-Paul 2003; Driver and Muñoz-Bugarin 2010; Hémous and Olsen 2022; Raurich, Sala, and Sorolla 2012). In the past four decades, technological change embodied in ICT capital has been biased

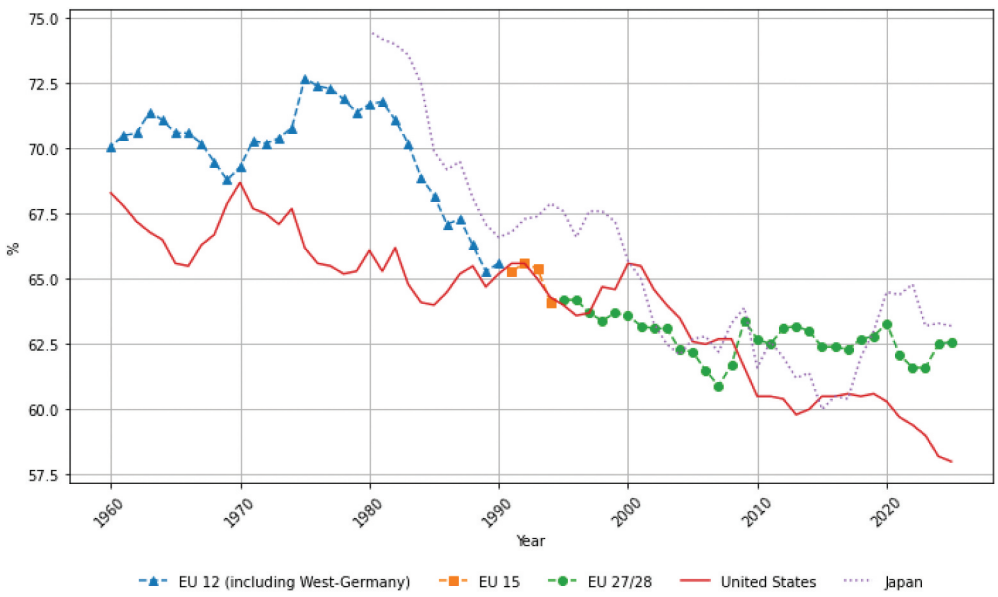

Figure 1. Dynamics of the wage share between 1960 and 2024. Source: Our own elaboration using the AMECO database. Note: As in the Quarterly Report on the Euro Area Vol. 18, n. 4 from 2018, the labour share is understood as the adjusted wage share present in the variable ALCD2 from AMECO: Compensation per employee as percentage of GDP at factor cost per person employed.

against the less educated (Frey and Osborne 2017),[9] with disembodied technical change biased towards high-skilled labour (Bassanini and Manfredi 2012). However, unlike past automation, the changes that ICT and in particular AI bring along affect also workers with higher education and skills (Freeman 2015), with college-educated individuals being more exposed, but also possibly better positioned to reap AI benefits (Cazzaniga et al. 2024). Moreover, the shift of income away from labour – and, in particular, away from low-wage workers – towards capital and top earners appears to have a negative impact on aggregate demand; workers receiving below-average pay tend to have a higher propensity to consume than top earners and capitalists (Dao 2020; Koo and Song 2016). However, empirical evidence concerning how different vocational groups are affected is yet inconclusive, with a literature review by Aghion et al. (2022) pointing to mixed results.

3. 'Re-shoring' – bringing back the jobs?

This development, combined with a rising geostrategic rivalry of Western countries and China, has unanticipated implications with respect to the relocation of production sites in the context of globalisation. Large international ICT companies, e.g. Apple, GE, IBM and General Motors, have moved some of their production facilities back to the United States and continue to do so. The Biden administration's Inflation Reduction Act linked some of the tax credits granted for private investments to local content requirements (LCRs), e.g. in the automotive and battery sectors (Scheinert 2023). How to bring industrial jobs back to Europe is also debated in the EU with a focus on critical sectors such as healthcare, semiconductors and energy (Raza et al. 2021). Patriotic sentiments or political considerations may also have a role in this development, but the fundamental drivers behind decisions to 'repatriate' jobs are economic (The Economist 2013). First, the labour portion of the cost of manufacturing – in particular in the ICT sector – has become so small that it is no longer a significant factor.[10] Second, the rising cost of transport (Ferrari, Christidis, and Bolsi 2023; World Bank 2021) and the risk of long supply chains (Chen 2018; Guerrieri et al. 2022) outweighs the remaining advantages of low-priced labour, making production in countries with higher priced labour profitable once again. Third, the costs of robots are (almost) the same everywhere.[11] Such 're-shoring' is expected to boost employment, if at all, mainly for high-quality, well-educated workers necessary to install and maintain machines and robots (The Economist 2013).

However, it should be stressed that substituting machine input for labour input is not deliberately intended to eliminate jobs. The purpose is to reduce costs. Less efficient machines are replaced by more efficient ones, production procedures are redesigned and new materials replace older ones – all for the purpose of lowering the costs of production. The catchphrase '*man vs. machine*' is misleading in that it assumes that human labour is still the main source of productive power. That is no longer true. Labour is a cost. Manufacture is driven by the need to reduce costs to make products cheaper and more affordable, thereby increasing sales and profits. The consequences of this central purpose on the free market, employment, purchasing power, income distribution, etc. – are secondary effects which have consequences of their own. But it is wrong to assume that these consequences are intended. After quality improvement, labour cost reduction is the most important motivation of manufacturing businesses to adapt automation-related technologies in the US (Dinlersoz and Wolf 2024, 611). Cost is the target.

Europe is also a contestant in the race for the machine. In June 2014, the European Commission and 180 companies and research organisations (under the umbrella of https://eu-robotics.net/) launched the world's largest civilian research and innovation programme in robotics (European Commission 2014). Although the programme promises, 'to create over 240,000 jobs in Europe, and increase Europe's share of the global market to 42%', it also stresses as one of its main goals the enabling of 'companies to continue manufacturing in Europe, where they might otherwise move operations to lower-cost countries'.

4. The elephant in the room

What deserves further investigation is the peculiar disconnect between the phenomena just described and public discussion and interpretation of their social and economic effects as manifested in economic studies, government reports, journalists' commentary, changes in corporate strategy, labour policy, legislation, etc.

- On the one hand, the discussion of automation is focused on its probable effects on labour employment, while the micro- and macro-distributive effects of this shift in productive roles are barely mentioned.
- On the other hand, the discussion of rising inequality is focused on issues of distributive justice; the shift from labour to capital as an income source and its active role in concentrating wealth is mostly ignored.

Thus, in a labour-focused environment, the distributive effects of automation which are a prime cause of expanding capital concentration are the 'elephant in the room' – an enormously portentous subject almost entirely omitted from public discussion. An exception is Freeman (2015) who argues in 'Who owns the robots rules the world', that *'[...] capital is a substantial contributor to inequality [...] in labor incomes because highly paid chief executive officers (CEOs) and top executives are paid stock options, restricted stock grants, and bonuses tied to capital income'* and that *'If we owned our replacements, we would have our current earnings and our time freed from labor to spend as we wished [...] If other persons owned our replacement robots, we would be jobless and searching for new work at lower pay while the owners of the robots would reap the pay/marginal product from the machines that took our jobs'*.

This omission is even more astounding since the 'elephant' is by no means a new or modern phenomenon: it goes back to the beginning of the Industrial Revolution,[12] which changed the proportional relationship between the input factors in favour of capital. What starts out as a perceived rise in labour productivity collapses in labour redundancy at the moment when production is fully automated. Instead of labour income for the many, its income yield ends up in the hands of a few – as an economic effect of concentrated ownership of productive capital.[13] At the same time, the argument that capital concentration is economically dysfunctional (Solimano 2017) is undisputed, as is the thesis that democracy requires a broad distribution of wealth (Wilkinson and Pickett 2010). Social attention, so far, however, has been focused on the growing wealth of the few (e.g. anti-monopoly legislation) without acknowledging the corresponding increase

in the number of those who do not own. The crucial point here is not the antagonistic relation between labour and capital, but rather the interaction of these factors. If ownership of the machines which are replacing labour in production were more widely spread among the population, working people could not only reduce their effective working hours, thus enhancing their 'productivity' while increasing their leisure, but at the same time replace their lost labour income (Rotman 2015).

The same blind spot is found in the heated discussion on inequality (Chancel et al. 2021, 10): Globally, the 10% wealthiest households own 76% of the global wealth. The middle 40% own 22% of the total, while the bottom 50% own not more than 2% of the global wealth in 2021. While the richest 10% earn 52% of global income, the bottom 50% earn 8.5%. Furthermore, due to an uneven recovery from the Great Recession, income inequality continued to grow even in the aftermath of the financial crisis between 2009 and 2015 (Saez 2018), a finding contrary to claims that crises are the big equaliser (see also Scheidel 2018). At the same time, the German philosopher Peter Sloterdijk (2009) complained that '*a good half of the population of every modern nation is made up of people with little or no income, who are exempt from taxes and live, to a large extent, off the other half of the population, which pays taxes*'.[14] And in fact, for example, 2023 income tax data from Italy show, 63.4% of income tax revenue is paid by 15.3% of the taxpayers, while 45% declare no income at all (www.finanze.gov.it/it/statistiche-fiscali/). In Sloterdijk's view, the unproductive increasingly live at the expense of the productive. However, he does not ask why half of the population is economically dependent on charity from the other half nor does he show any interest in the cause of this occurrence.

One of the few analyses taking into account historical data to assess the distributional effects of measures branded as antidote, in this case employee financial participation (EFP), were published by J. Blasi and Kruse (2024). In the first society-wide measure of the impact of EFP on one economy, i.e. the US, they show that, put into the context of the concentration of capital ownership and the concentration of capital income between 1979 and 2018, EFP is declining in terms of its proportion in household income.[15] In light of automation and the rise of AI, Acemoglu, Manera, and Restrepo (2020) argue that the US tax system is biased against labour and in favour of capital. Taxing labour heavily while having low tax rates on capital investments in technology provides strong incentives for firms to automate employment.

5. 'Creative destruction' of jobs – the interim balance in industrial societies 2022

The distributive effects discussed above would be less drastic, of course, if employment were destroyed and created at the same rate, and if the new jobs were qualitatively the same as the old ones. However, Brynjolfsson and McAfee (2011, 2014) and McAfee and Brynjolfsson (2018) argue that this is not the case. Technological change in the twenty-first century is both faster and more widespread than in the past as computers and ICT, unlike the steam engine, electric motor or internal combustion engine, are subject to continuous improvement.[16] They provide evidence for divergences between higher-skilled and lower-skilled workers, between 'superstars' and average workers, as well as between capital and labour, a process that – as they say – is inclined to leave more and

more people behind, although benefiting the economy as a whole (Brynjolfsson and McAfee 2011).

Another scenario mitigating the negative distributive effects mentioned would be an EU non-growth policy. However, in regard to the relationship between employment and productivity, it seems unlikely that a policy of reindustrialisation, still on the European agenda (European Commission 2020; Raza et al. 2021), will leave much room for the intentional non-utilisation of the potential for increasing productivity that modern ICT offers. On the contrary, policy-makers can be expected to exploit this potential in order to exit the great recession. A comparison of the annual growth rates in the US and Europe (see Table 1) well illustrates this trade-off.

Table 1. Employment vs. Productivity in the US and Europe (Euro Area).

Economic Region	1970–1980	1980–1990	1990–2000	2000–2010	2010–2020
United States			Annual growth in %		
Employment	2.14	1.72	1.30	0.33	1.24
Productivity	1.03	1.51	2.09	1.62	1.20
Output	3.17	3.23	3.40	1.95	2.43
Europe (Euro Area)*			Annual growth in %		
Employment	0.47	0.54	0.33	0.90	0.63
Productivity	2.93	1.56	1.60	0.57	−0.13
Output	3.40	2.10	1.93	1.47	0.50

Employment: Growth in employment for the United States and for the Euro Area. By Euro Area Average growth rates per decade. **Productivity**: Growth in labor productivity per person employed for the United States and the Euro Area. Average growth rates per decade. **Output**: measured as in Landmann (2004), p. 21. * Historic data of the countries that now are EU Member States. **Source**: Own calculations based on the Conference Board Total Economy Database™, April 2023.

Between 1970 and 1980 growth in economic output was similar. During the next two decades in the US productivity growth more than doubled with decreasing employment growth maintaining overall output growth. During the same interval, with productivity declining and employment after a short moderate increase being also on the decline, overall output decreased in the EU. This trend was reversed in the US between 2000 and 2020, when regaining an increase in output rise was accompanied by decreased productivity growth and and an increase in employment growth. In the EU, after a recovery of employment growth productivity growth remained on the decline eventually turning negative resulting in a continuous decline in overall output growth. In other words, productivity increased at the cost of employment and vice versa. Interestingly, both economic regions show a similar decline in wage share over the past 40 years (see above Figure 1).

But even if job creation keeps pace with job destruction, the distributive effects of automation will still lead to increasing capital concentration. As illustrated in Figure 2, labour productivity as a measure of production output per human hour worked has become uncoupled from remuneration in the past: Between 1948 and 1979, the average annual growth rates for productivity and compensation (including that of top earners as well as unskilled workers) were similar, at 2.5% and 2.1%. Since then, the average annual growth rate for productivity was 1.4%, while compensation grew by only 0.6%. In summary, between 1979 and 2024 the average hourly compensation grew just 29.4%, lagging a productivity growth of 80.9% in the same interval, a phenomenon dubbed

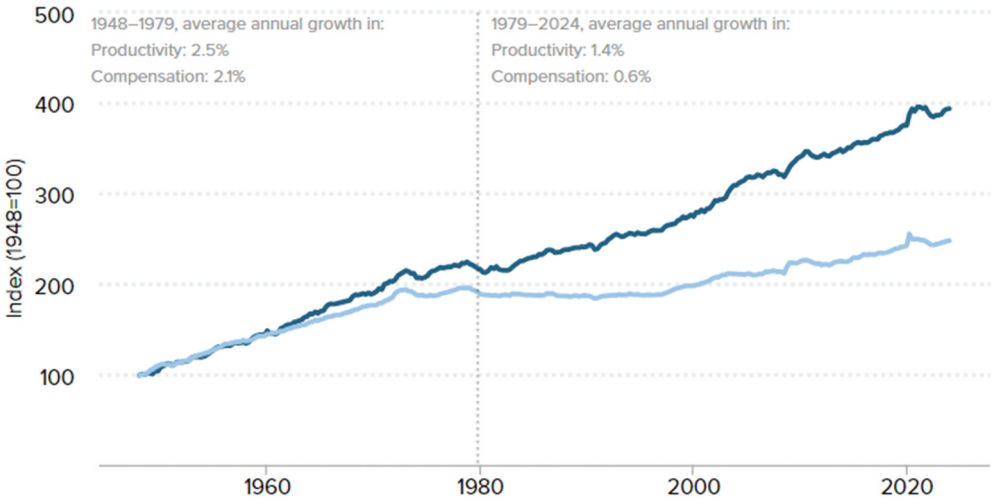

Figure 2. Productivity growth and hourly compensation growth, 1948–2024 (Bivens et al. (2014), updated in EPI (2024)). Data are for compensation (wages and benefits) of production/nonsupervisory workers in the private sector and net productivity of the total economy. 'Net productivity' is the growth of output of goods and services less depreciation per hour worked. **Source**: EPI analysis of unpublished Total Economy Productivity data from Bureau of Labor Statistics (BLS) Labor Productivity and Costs program, wage data from the BLS Current Employment Statistics, BLS Employment Cost Trends, BLS Consumer Price Index, and Bureau of Economic Analysis National Income and Product Accounts Updated from Figure A in *Raising America's Pay: Why It's Our Central Economic Policy Challenge* (Bivens et al. 2014).

'gross decoupling'. Critical voices (Pessoa and Van Reenen, 2013) argue, though, that when calculating this rate in terms of net decoupling, the effect decreases when including increased non-wage labour costs, e.g. pension contributions, healthcare benefits and other factors. However, they explain the difference of gross and net decoupling with two key factors, namely inequality and non-wage labour cost, while, nonetheless, acknowledging that increasing wage inequality is an important factor contributing to the phenomenon of 'decoupling'. In conclusion, most recent research (Acemoglu and Restrepo, 2022) corroborates the findings of Bivens et al. ((2014) updated 2024 in Figure 2), in particular with regard to the effect of wage inequality and its roots in the shift between the factors of production, which is most relevant in the context of this article. Further, Tyson and Zysman (2022) argue accurately that if more and more jobs were automated, even less of the mentioned non-wage labour costs like pension contributions and healthcare benefits were to be paid.

It is predictable that in tomorrow's world productivity will still be rising sharply but this increase will become increasingly disconnected from wages and salaries for the many, while the few will benefit from ever steeper increases in executive remuneration and capital earnings.[17] As technological progress embodied in ICT capital has been biased against the less educated and the average worker (and only in some cases towards high-skilled labour) – as argued earlier – concomitant changes in the employment structure will worsen the negative distributive effects. However, as the digital revolution generally reduces labour input across the board in an economy based on growth, labour

income falls as capital-based incomes rise; consequently, wealth inequality will continue to increase. These findings are corroborated by both J. Blasi and Kruse's (2024) analysis of distributional effects of EFP and the work of Skare, Gavurova, and Burić (2024) on AI and wealth inequality, revealing a positive, statistically significant correlation between AI technology adoption, AI capital stock accumulation and wealth disparity.

6. The postulate: extending financial participation to broader constituencies

To offset the described disadvantages of AI/automation, mitigate and – if possible – reduce further capital concentration, all citizens should be enabled to (co-)own the technologies increasingly producing the wealth of our post-industrial economies and benefit from capital ownership providing them with a second income source independent of labour income.

The Polish writer and futurist *Stanisław Lem*, who took a great interest in cybernetics and robotics,[18] anticipated these problems in Poland as early as 1954 (Lem 1954). Four years later, in 1958, the American corporate lawyer and merchant banker *Louis O. Kelso* presented an analysis of the fundamental causes of this phenomenon and a proposal for dealing with its economic effects in his book 'The Capitalist Manifesto' (Kelso and Adler 1958; see also Kelso and Hetter 1967). He invented a novel financial mechanism, the prototype of the leveraged buyout, to finance ownership of productive property for employees, consumers and the general population. As mentioned earlier, Kelso is best known as the originator of the ESOP – Employee Stock Ownership Plan – in which capital ownership of thousands of companies has been diffused to each company's employees. But he also implemented less common financial tools as the Consumer Stock Ownership Plan (SOP) or the General Stock Ownership Plan (GSOP). The difference to the ESOP is the group of plan participants: In ESOPs it is employees, in CSOPs consumers of a utility and in GSOPs citizens of a region. All plans use the same financing logic that employs an intermediary corporate vehicle, facilitates the involvement of individual investors through a trusteeship and may use external financing, thereby achieving the benefit of financial leverage. To overcome the problem that the majority of citizens is deprived of savings or access to capital credit, Kelso's plans use the borrowing power of the corporate entity to finance the purchase of shares in that company by the different constituencies; the acquisition loan is repaid from the future earnings of the credit-financed shares.[19]

While Neelie Kroes, Vice President of the European Commission, asserted in 2014 that '*Europe needs to be a producer and not merely a consumer of robots*', Kelso would postulate a 'race for the machine' to enable citizens to become owners of the robots who both serve and replace them. The root of the problem, Kelso argued, was access to capital credit in order to acquire productive property, in other words the non-human things that produce wealth in an industrial society.[20] A solution to the dilemma of machines substituting for humans and reducing labour demand would be for European citizens to acquire a significant share of (co-)ownership in the robots competing with them on the labour market, subsequently providing them with a second source of income independent of their labour (Freeman 2015). This is in line with the support of the European

Commission for Employee Ownership (Lowitzsch et al. 2014) and the findings of the most recent PEPPER V Report (Lowitzsch and Hashi 2024) looking back on five decades of EFP across the European Union and showing that the principal mechanisms of company law, corporate governance, fiscal and other incentives are mostly identical for both employee and consumer financial participation and that ample synergies exist. Intermediary entities that facilitate consumer (co-)ownership are similarly diverse, with RE cooperatives, LLCs, foundations and associations being used just as in the realm of employee share ownership (Lowitzsch 2019b).

Kelso called such types of financial investment 'universal capitalism' – the idea that *everyone* should be able to benefit from ownership of productive assets, instead of just a tiny wealthy sliver who can afford the risk. Financial participation with its acknowledged positive effects on social cohesion and motivation (J. R. Blasi, Kruse, and Bernstein 2003) should not be limited to employees but extended as a focused policy also to other constituencies, for example, to citizens as consumers.

a) Best Practice: The Energy Transition and the 'Proximity and Social Economy industrial ecosystem'

A best practice example for how to legislate said extension of financial participation was the passing of the 'Clean Energy Package' (CEP) – a comprehensive legislative overhaul of the European Energy Union – in winter 2018/19. As part of the CEP, the European Union passed a legal framework for consumer (co-)ownership, i.e. prosumership,[21] in the recast of the Renewable Energy Directive (RED II) and in the Internal Electricity Market Directive (IEMD). Energy communities, mentioned and defined in both directives,[22] have the explicit aim of providing environmental, economic, or social community benefits for their members, but may be profit making. Consumers, as prosumers, now have the right to consume, store or sell RE generated on their premises (a) individually, for example households and micro enterprises, or jointly acting, for example in tenant electricity projects; and (b) collectively as part of energy communities organised as independent legal entities. As a result, there is a growing number of energy communities involving an increasing share of (co-)ownership of broader strata of European populations (Lowitzsch 2019a); the EC estimates 9,000 energy communities currently in operation across the EU.[23] In contrast to EFP – characterised by an absence of European legislation – binding rules for consumer (co-)ownership have already been transposed in the majority of the 27 EU Member States (Hoicka et al. 2021).

An interlinked policy area lending itself to extending financial participation to broader strata of society is the 'Proximity and Social Economy industrial ecosystem'. Since the 2011 Social Business Initiative (SBI) and the 2016 Start-up and Scale-up initiative, the European Commission has postulated to harness the economic potential of social enterprises in view of their crucial role in fostering sustainable economic growth. Social enterprises have been present for decades delivering innovative green solutions, for example, in the circular economy.[24] Presented in the framework of the EU Industrial Strategy, the EC has launched a new initiative for the beginning of 2022 to boost the social economy contribution to the green transition as captured under the heading *'The social economy business case for the green transition'* (European Commission 2021).[25]

Acknowledging the strong presence and pioneering role of social enterprises in climate change mitigation and adaptation, clean energy transition, circular economy and related areas, this policy action directly contributes to developing the corresponding institutional environment of the 'Proximity and Social Economy industrial ecosystem' (European Commission 2022).

Of course, the renewable energy sector and the 'Proximity and Social Economy industrial ecosystem' are just examples of recent successful policy initiatives, and the broadening of financial participation to other constituencies should be considered and supported.

b) Examples of business models involving employee and consumer co-ownership

Such an undertaking, other than political support, requires appropriate business models that are flexible enough to involve employees and consumers as key stakeholders of this SME environment. The following two examples illustrate that there are no legal barriers to combine employee and consumer financial participation, while stressing the potential of such business models.

- The Renewable Energy Consumer Stock Ownership Plan (RE-CSOP), as implemented in the energy world, provides a governance model for heterogeneous co-investments of municipalities, citizens and SMEs in (renewable) energy communities (Lowitzsch, Hoicka, and Van Tulder 2020). CSOPs were pioneered in the Horizon 2020 project SCORE,[26] which ran from 2018 to 2021 in five pilot regions and in cities across Europe following these pilot projects (Lowitzsch 2022).[27] SCORE reached out to more than 700 local authorities and 10,000 consumers demonstrating the positive impact (co-)ownership has on consumer behaviour and showing the ability of this democratic participation model to include under-represented groups. Again, the energy sector is just prototypical, and CSOPs can be employed also in other settings such as investments in climate warming mitigation measures.
- The French Société coopérative d'intérêt collectif (SCIC, Cooperative society of collective interest)[28] can take the form of a public limited company, a simplified joint stock company or a limited liability company. Anchored in a geographical area or professional community the SCIC, organised around a common economic activity or interest, can cover any type of activity that provides services to organisations or individuals. To legally qualify, a SCIC must involve (i) employees, farmers, craftsmen, or the like; (ii) beneficiaries, i.e. customers, suppliers, volunteers, etc.; and (iii) depending on its aims, private companies, financiers, associations, or local authorities, with the latter being allowed to hold up to 50% of the equity.

c) Outlook: A revised research agenda

The discussion on democratising and broadening ownership of productive assets seems to transcend ideological grounds and more and more centres on questions of how to increase cohesion among our societies and how to cope with the new challenges, for example climate warming and the Energy Transition, most efficiently. At the same time,

policy-makers are increasingly perceptive to arguments supporting employee and consumer (co-)ownership and have begun to understand that the behavioural changes necessary to preserve our habitat and make our economies more equitable require corresponding incentives. Regarding (co-)ownership in robots, co-bots and AI, more research is needed to shed light, not so much on labour market effects but on questions of distribution and capital concentration. Responding to such need, in 2019 the European Commission set up a European *AI on Demand-Platform* (AI4EU) with the aim to '*create a thriving European AI research ecosystem driven by AI excellence, through a channel that fosters collaboration, reproducibility and experimentation, while maximising academic, social and industrial impact*'.[29]

The potential for (co-)ownership of employees in the enterprises they work for, and of consumers in the utilities they are served by, should inspire each other and possibly be harnessed under a joint approach. This is particularly important since as Acemoglu and Johnson (2023) conclude in '*Power and progress*' the prosperity of our societies was not an automatic gain of technological progress, but the result of conscientious decisions to share the results of innovation with broader strata of society instead reserving them for a narrow elite.

Notes

1. The three plans are variations of a capital credit device for utilising corporate credit to simultaneously finance both corporate growth and asset acquisition. The plan enables employees (or consumers or citizens respectively) to buy stock in a corporation and to pay for it out of that stock's future earnings. As of 2021 there were about 6,533 ESOPs with more than 14.7 million employee participants in the U.S.
2. See the article by K. Schwab who coined the term: https://www.weforum.org/agenda/2016/01/the-fourth-industrial-revolution-what-it-means-and-how-to-respond..
3. A.I. has become established in tech, manufacturing, healthcare, banking and financial services, media, retail, hospitality, and automaking. Its fast pace is exemplified, e.g. by ChatGPT within two months of its initial release had 100 million monthly active users around the globe, see R. Stropoli, 'A.I. Is Going to Disrupt the Labor Market. It Doesn't Have to Destroy It'. CBR – Artificial Intelligence, 14 Nov., 2023.
4. O.J. L 2024/1689 of 12 July 2024; after passing the European Parliament on 13 March 2024 (a vote of 523 for, 46 against, and 49 abstaining) the law was approved by the EU Council on 21 May and entered into force on 1 August 2024.
5. Costs for the humanoid robot created by Agility Robotics, used by Amazon in its warehouse operations in the U.S. are in US Dollars $10 to $12. Projections expect the costs to decrease, eventually reaching between 2 and 3 EUR per hour. See Kirschner, *Business Insider*, 6 Dec 2023: Amazon's humanoid warehouse robots will eventually cost only $3 per hour to operate.
6. Related proposals were discussed in many countries worldwide, for example, in the French, British and German parliament (see Die Zeit 'Ab zum Finanzamt Kollege', No 31 of 21 July 2016).
7. P. Kelly, The Guardian, 12 May 2023, '"Why would we employ people?" Experts on five ways AI will change work'.
8. See also J. Ma, Fortune, 25 May 2024, 'Top VC Kai-Fu Lee says his prediction that AI will displace 50% of jobs by 2027 is "uncannily accurate"'.
9. See also (Dengler and Matthes 2015, 6) and *Evans-Pritchard, Ambrose* 'Robots may shatter the global economic order within a decade', the Telegraph 5 Nov. 2015.

10. 'Welcome home – The outsourcing of jobs to faraway places is on the wane. But this will not solve the West's employment woes', The Economist, 19 Jan. 2013.
11. Average cost of industrial robots dropped from ~69,000 USD in 2005 to ~27,000 USD in 2017; the prognosis for the average price in 2025 is 10,900 USD (for the figures see Ark Invest 2019 at https://www.ark-invest.com/articles/analyst-research/industrial-robot-cost-declines).
12. See, e.g. the 'Leeds Woollen Workers Petition' of 1786 complaining about the effects of machines on the previously well-paid skilled workers; http://www.fordham.edu/halsall/mod/1786machines.asp.
13. A good example is given by David Rotman in 'How Technology Is Destroying Jobs', MIT Review of June 12, 2013: *'someone who creates a computer program to automate tax preparation might earn millions or billions of dollars while eliminating the need for countless accountants'*.
14. 'Die Revolution der gebenden Hand' (Revolution of the giving Hand), FAZ 13 June 2009; the article kicked off a long controversy in German media, which, however, did not address the shift between labour and capital.
15. Capital income has largely collapsed between 1979 and 2018 for the working middle class. 59.3% of all capital income was received by the richest 1% of households in 2018 (compared with 39.6% in 1979); 89.7% of all capital income was received by the richest 20% of households in 2018 (compared with 76.2% in 1979).
16. Moor´s law (1965) predicts that computers' abilities as an extension of their processing abilities would double every 12 months, whereas today the pace of increase in computing capacity is believed to be even faster.
17. See John Lanchester 'The Robots Are Coming' (Lanchester 2015) who also gives an illustrative example for that trend: '*In 1960, the most profitable company in the world's biggest economy was General Motors. In today's money* [i.e. as of 2015], *GM made $7.6 billion that year. It also employed 600,000 people. Today's most profitable private-owned company [i.e. Apple] employs 92,600. So where 600,000 workers would once generate $81 billion in profit, now 92,600 generate $89.9 billion, an improvement in profitability per worker of* 76.65 *times'*.
18. The word robot was coined by the Czech interwar writer *Karel Čapek* in his play R.U.R. (Rossum's Universal Robots), published in 1920. Literal meaning of the word 'robota' is 'corvée', 'serf labor', and figuratively, 'drudgery' or 'hard work' in Czech; also (more general) 'work', 'labour', in many Slavic languages.
19. An ESOP, for example, ordinarily involves a loan to an employee benefit trust, which acquires company stock and allocates it through periodic contributions to each employee's ESOP account. The loan may be serviced by payments by the company from company profits or from dividends paid on the ESOP stock.
20. It is worth noting here that the Polish trade union 'Solidarity' celebrated Kelso's visit to Warsaw in October 1989, and insisted on making employee ownership a part of the privatisation process. 'Solidarity' took a quite different position from that of most European trade unions at the time.
21. As early as 1972 Marshall McLuhan and Barrington Nevitt suggested in their book *Take Today*, (p. 4) that technological progress would transform the consumer into a producer of electricity. The artificial word stemming from the Latin was probably first introduced by Alvin Toffler in his book *The Third Wave* (1980).
22. While the RED II focuses on the promotion of renewable energies (RE) and thus speaks of 'Renewable Energy Communities' (RECs) the IEMD as the more general legal act addresses 'Citizen Energy Communities' (CECs).
23. https://energy.ec.europa.eu/news/focus-energy-communities-transform-eus-energy-system-2022-12-13_en.
24. Examples are providing re-use and up- and recycling services, generating collaborative economy models, eco-design, but also RE production as the already large and dynamically growing population of RECs and CECs demonstrate (European Commission 2022).

25. https://single-market-economy.ec.europa.eu/sectors/proximity-and-social-economy/transition-pathway_en.
26. SCORE. Horizon 2020 SCORE (GA No. 784960) n.d. Available online: https://kelso-institute-europe.de/projects/score/.
27. SCORE reached out to more than 700 local authorities and 10,000 consumers demonstrating the positive impact co-ownership has on consumer behaviour and showing the ability of this democratic participation model to include underrepresented groups.
28. https://www.les-scic.coop/chiffres-cles.
29. https://aiod.eu/about.

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References

Acemoglu, D., and S. Johnson. 2023. *Power and Progress: Our Thousand-Year Struggle Over Technology and Prosperity*. New York: PublicAffairs.

Acemoglu, D., A. Manera, and P. Restrepo. 2020. "Does the US Tax Code Favor Automation?" *Working Paper*. National Bureau of Economic Research. https://doi.org/10.3386/w27052

Acemoglu, D., and P. Restrepo. 2020. "Robots and Jobs: Evidence from US Labor Markets." *Journal of Political Economy* 128 (6): 2188–2244. https://doi.org/10.1086/705716.

Acemoglu D and Restrepo P. (2022). Tasks, Automation, and the Rise in U.S. Wage Inequality. *ECTA*, 90 (5): 1973–2016. https://doi.org/10.3982/ECTA19815.

Aghion, P., C. Antonin, S. Bunel, and X. Jaravel. 2022. "The Effects of Automation on Labor Demand." In *Robots and AI*. L. Y. Ing and G. Grossman, (Eds), 15–39. 1st ed. Routledge. https://doi.org/10.4324/9781003275534-2.

Arpaia, A., E. Pérez, and K. Pichelmann. 2009. "Understanding Labour Income Share Dynamics in Europe." https://doi.org/10.2765/37596.

Bassanini, A., and T. Manfredi. 2012. *Capital's Grabbing Hand? A Cross-Country/cross-Industry Analysis of the Decline of the Labour Share*. Vol. 133. OECD. https://doi.org/10.1787/5k95zqsf4bxt-en.

Bentolila, S., and G. Saint-Paul. 2003. "Explaining Movements in the Labor Share." *Contributions in Macroeconomics* 3 (1). https://doi.org/10.2202/1534-6005.1103.

Bivens, J., E. Gould, L. Mishel, and H. Shierholz. 2014. "Raising America's Pay. Why It's Our Central Economic Policy Challenge." *Briefing Paper No. 378*. Ecnomic Policy Institute.

Blasi, J., and D. Kruse. 2024. "Employee Financial Participation and the Rising Concentration of Capital Ownership and of Capital Income." *Journal of Participation and Employee Ownership* 7 (2): 138–154. https://doi.org/10.1108/JPEO-06-2024-0009.

Blasi, J. R., D. L. Kruse, and A. Bernstein. 2003. *In the Company of Owners: The Truth About Stock Options* (And Why Every Employee Should Have Them). New York: Basic Books.

Brynjolfsson, E., and A. McAfee. 2011. *Race Against the Machine: How the Digital Revolution is Accelerating Innovation, Driving Productivity, and Irreversibly Transforming Employment and the Economy*. Lexington: Digital Frontier Press.

Brynjolfsson, E., and A. McAfee. 2014. *The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies*. 1st ed. New York: W. W. Norton & Company.

Carbonara, E., C. Focacci, and E. Santarelli. 2023. "Mitigating the Labor Displacing Effects of Automation Through a Robot Tax: Evidence from a Survey Experiment." *Economics of Innovation & New Technology* 33 (8): 1145–1158. https://doi.org/10.1080/10438599.2023.2293031.

Cazzaniga, M., F. Jaumotte, L. Li, G. Melina, A. Panton, C. Pizzinelli, and E. Rockall. 2024. *Gen-AI: Artificial Intelligence and the Future of Work*. International Monetary Fund. https://doi.org/10.5089/9798400262548.006.

Chancel, L., T. Piketty, E. Saez, and G. Zucman. 2021. *World Inequality Report 2022. World Inequality Lab*. https://wir2022.wid.world/www-site/uploads/2023/03/D_FINAL_WIL_RIM_RAPPORT_2303.pdf.

Chen, H. L. 2018. "Supply Chain risk's Impact on Corporate Financial Performance." *International Journal of Operations & Production Management* 38 (3): 713–731. https://doi.org/10.1108/IJOPM-02-2016-0060.

Dao, M. 2020. "Wealth Inequality and Private Savings: The Case of Germany." *IMF Working Papers*, *20*(107). https://doi.org/10.5089/9781513546063.001.

Dengler, K., and B. Matthes. 2015. *In kaum einem Beruf ist der Mensch vollständig ersetzbar* (IAB-Kurzbericht No. 24/2015). Institut für Arbeitsmarkt- und Berufsforschung. https://doku.iab.de/kurzber/2015/kb2415.pdf.

Dinlersoz, E., and Z. Wolf. 2024. "Automation, Labor Share, and Productivity: Plant-Level Evidence from U.S. Manufacturing." *Economics of Innovation & New Technology* 33 (4): 604–626. https://doi.org/10.1080/10438599.2023.2233081.

Driver, C., and J. Muñoz-Bugarin. 2010. "Capital Investment and Unemployment in Europe: Neutrality or Not?" *Journal of Macroeconomics* 32 (1): 492–496. https://doi.org/10.1016/j.jmacro.2009.03.003.

The Economist. 2013. Here, There and Everywhere. *Special Report. Outsourcing and Offshoring*. https://www.economist.com/special-report/2013/01/17/here-there-and-everywhere.

EPI (Economic Policy Institute). 2024. *The Productivity–Pay Gap*. https://www.epi.org/productivity-pay-gap/.

European Commission. 2014. *EU Launches world's Largest Civilian Robotics Programme – 240,000 New Jobs Expected* (Press Release No. IP/14/619). https://ec.europa.eu/commission/presscorner/detail/en/ip_14_619.

European Commission. 2020. *A New Industrial Strategy for Europe* (Communication No. COM(2020) 102 final). https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52020DC0102.

European Commission. 2021. *Scenarios Towards Co-Creation of a Transition Pathway for a More Resilient, Sustainable and Digital Proximity and Social Economy Industrial Ecosystem* (Commission Staff Working Document No. SWD(2021) 982 final). https://ec.europa.eu/docsroom/documents/47854.

European Commission. 2022. *Transition Pathway for Proximity and Social Economy Ecosystem*. Publications Office. https://doi.org/10.2873/44306.

European Parliament. 2017. "European Parliament Resolution of 16 February 2017 with Recommendations to the Commission on Civil Law Rules on Robotics." *Pub L No (2015/2103 (INL)* C:252 OJ 239. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52017IP0051.

European Union Regulation (EU) 2024/1689, L OJ. 2024. https://eur-lex.europa.eu/eli/reg/2024/1689/oj.

Federal Statistical Office. 2023. *One Hour Worked Cost an Average of 39.50 Euros in 2022*. April 26. https://www.destatis.de/EN/Press/2023/04/PE23_164_624.html.

Ferrari, E., P. Christidis, and P. Bolsi. 2023. "The Impact of Rising Maritime Transport Costs on International Trade: Estimation Using a Multi-Region General Equilibrium Model." *Transportation Research Interdisciplinary Perspectives* 22:100985. https://doi.org/10.1016/j.trip.2023.100985.

Freeman, R. 2015. "Who Owns the Robots Rules the World." *IZA World of Labor* 5. https://doi.org/10.15185/izawol.5.

Frey, C., and M. Osborne. 2017. "The Future of Employment: How Susceptible are Jobs to Computerisation?" *Technological Forecasting & Social Change* 114:254–280. https://doi.org/10.1016/j.techfore.2016.08.019.

Guerrieri, V., G. Lorenzoni, L. Straub, and I. Werning. 2022. "Macroeconomic Implications of COVID-19: Can Negative Supply Shocks Cause Demand Shortages?" *The American Economic Review* 112 (5): 1437–1474. https://doi.org/10.1257/aer.20201063.

Hall, R., R. Ashford, N. Ashford, and J. Arango-Quiroga. 2019. "Universal Basic Income and Inclusive Capitalism: Consequences for Sustainability." *Sustainability* 11 (16): 4481. https://doi.org/10.3390/su11164481.

Hémous, D., and M. Olsen. 2022. "The Rise of the Machines: Automation, Horizontal Innovation, and Income Inequality." *American Economic Journal Macroeconomics* 14 (1): 179–223. https://doi.org/10.1257/mac.20160164.

Hoicka, C., J. Lowitzsch, M. C. Brisbois, A. Kumar, and L. Ramirez Camargo. 2021. "Implementing a Just Renewable Energy Transition: Policy Advice for Transposing the New European Rules for Renewable Energy Communities." *Energy Policy* 156:112435. https://doi.org/10.1016/j.enpol.2021.112435.

Holzer, H. 2021. "After COVID-19: Building a More Coherent and Effective Workforce Development System in the US." *Policy Paper*). IZA Institute of Labor Economics. https://docs.iza.org/pp174.pdf

Kelso, L., and M. Adler. 1958. *The Capitalist Manifesto*. New York: Random House.

Kelso, L., and P. Hetter. 1967. *Two-Factor Theory: The Economics of Reality*. New York: Random House.

Koo, J., and Y. Song. 2016. "The Relationship Between Income Inequality and Aggregate Saving: An Empirical Analysis Using Cross-Country Panel Data." *Applied Economics* 48 (10): 892–901. https://doi.org/10.1080/00036846.2015.1090548.

Lanchester, J. 2015. "The Robots are Coming." *London Review of Books*, *37*(5). March 5. https://www.lrb.co.uk/the-paper/v37/n05/john-lanchester/the-robots-are-coming.

Lem, S. 1954. *Dzienniki gwiazdowe—Podróż dwudziesta czwarta Ijona Tichego (The Star Diaries—The 24th Voyage of Ijon Tichy)*. Wydawnictwo Iskry.

Lowitzsch, J. 2019a. "Consumer Stock Ownership Plans (CSOPs)—The Prototype Business Model for Renewable Energy Communities." *Energies (Basel)* 13 (1): 118. https://doi.org/10.3390/en13010118.

Lowitzsch, J., Ed. 2019b. *Energy Transition: Financing Consumer Co-Ownership in Renewables*. Springer International Publishing. https://doi.org/10.1007/978-3-319-93518-8.

Lowitzsch, J. 2022. H2020 Final Report. Consumer Stock Ownership Plans (CSOPs)—Financing Energy Communities. Europa-Universität Viadrina Frankfurt (Oder) & Kelso Institute Europe. https://csd.eu/fileadmin/user_upload/publications_library/files/2022_04/SCORE_Final_Report_-_Web_V_2022.pdf .

Lowitzsch, J., I. Hashi. 2014. *Study on the Promotion of Employee Ownership and Participation*. Brussels: Brussels; study forC DG MARKT.

Lowitzsch, J., and I. Hashi, Eds. 2024. *The PEPPER V Report. Benchmarking Employee Participation in Profits and Enterprise Results in the European Union, the United Kingdom and the United States*. Kelso Institute Europe. https://doi.org/10.11584/PEPPER-5.

Lowitzsch, J., C. Hoicka, and F. Van Tulder. 2020. "Renewable Energy Communities Under the 2019 European Clean Energy Package – Governance Model for the Energy Clusters of the Future?" *Renewable and Sustainable Energy Reviews* 122:109489. https://doi.org/10.1016/j.rser.2019.109489.

McAfee, A., and E. Brynjolfsson. 2018. *Machine, Platform, Crowd: Harnessing Our Digital Future*. New York: W. W. Norton & Company.

Pessoa, J. P., & Van Reenen, J. (2013). Decoupling of wage growth and productivity growth? Myth and reality. CEP Discussion Papers dp1246, Centre for Economic Performance, London School of Economics.

Piketty, T., and A. Goldhammer. 2014. *Capital in the Twenty-First Century*. Cambridge, Massachusetts: The Belknap Press of Harvard University Press.
Raurich, X., H. Sala, and V. Sorolla. 2012. "Factor Shares, the Price Markup, and the Elasticity of Substitution Between Capital and Labor." *Journal of Macroeconomics* 34 (1): 181–198. https://doi.org/10.1016/j.jmacro.2011.09.004.
Raza, W., J. Grumiller, H. Grohs, J. Essletzbichler, and N. Pintar. 2021. *Post COVID-19 Value Chains: Options for Reshoring Production Back to Europe in a Globalised Economy*. European Parliament.
Rifkin, J. 2004. *The End of Work: The Decline of the Global Work-Force and the Dawn of the Post-Market Era*. Updated paperback ed. New York: Jeremy P. Tarcher.
Rotman, D. 2015. "Who Will Own the Robots?" *MIT Technology Review*. https://www.technologyreview.com/2015/06/16/11184/who-will-own-the-robots/.
Saez, E. 2018. "Striking it Richer: The Evolution of Top Incomes in the United States." In *Inequality in the 21st Century: A Reader*, edited by D. Grusky and J. Hill, 39–42. New York: Routledge.
Scheidel, W. 2018. *The Great Leveler: Violence and the History of Inequality from the Stone Age to the Twenty-First Century*. Princeton University Press. https://doi.org/10.23943/9780691184319.
Scheinert, C. 2023. *Eu's Response to the US Inflation Reduction Act (IRA)* [Briefing]. Publications Office of the European Union. https://doi.org/10.2861/0467.
Skare, M., B. Gavurova, and S. B. Burić. 2024. "Artificial Intelligence and Wealth Inequality: A Comprehensive Empirical Exploration of Socioeconomic Implications." *Technology in Society* 79:102719. https://doi.org/10.1016/j.techsoc.2024.102719.
Sloterdijk, P. 2009. "Die Revolution der gebenden Hand." *Frankfurter Allgemeine Zeitung*. June 13. https://www.faz.net/aktuell/feuilleton/debatten/kapitalismus/die-zukunft-des-kapitalismus-8-die-revolution-der-gebenden-hand-1812362.html .
Solimano, A. 2017. *Global Capitalism in Disarray: Inequality, Debt, and Austerity*. New York: Oxford University Press.
Thuemmel, U. 2023. "Optimal Taxation of Robots." *Journal of the European Economic Association* 21 (3): 1154–1190. https://doi.org/10.1093/jeea/jvac062.
Tyson, L., and J. Zysman. 2022. "Automation, AI & Work." *Proceedings of the American Academy of Arts and Sciences* 151 (2): 256–271. https://doi.org/10.1162/daed_a_01914.
Wilkinson, R., and K. Pickett. 2010. *The Spirit Level: Why Equality is Better for Everyone*. London: Penguin Books.
World Bank. 2021. "High Trade Costs: Causes and Remedies." *Global Economic Prospects, June 2021*: 103–128. https://doi.org/10.1596/978-1-4648-2058-8.
World Economic Forum. 2023. Future of Jobs Report 2023 [Insight Report]. https://www3.weforum.org/docs/WEF_Future_of_Jobs_2023.pdf.

Appendix

Annex: Stanisław Lem excerpt from the 1954 **Memoirs of a Space Traveler**
Further Reminiscences of Ijon Tichy, The 24th Voyage:

. . .

" . . . Through the ages our inventors built machines that simplified work, and where in ancient times a hundred Drudgelings had bent their sweating backs, centuries later a few stood by a machine. Our scientists improved the machines, and the people rejoiced at this, but subsequent events showed how cruelly premature was that rejoicing. A certain learned constructor built the New Machines, devices so excellent that they could work quite independently, without supervision. And that was the beginning of the catastrophe. When the New Machines appeared in the factories, hordes of Drudgelings lost their jobs; and, receiving no salary, they faced starvation . . . "

"Excuse me, Phool," I asked, "but what became of the profits the factories made?"

"The profits," he replied, "went to the rightful owners, of course. Now, then, as I was saying, the threat of annihilation hung . . . " "But what are you saying, worthy Phool!" I cried. "All that had to be done was to make the factories common property, and the New Machines would have become a blessing to you!"

The minute I said this the Phool trembled, blinked his ten eyes nervously, and cupped his ears to ascertain whether any of his companions milling about the stairs had overheard my remark.

"By the Ten Noses of the Phoo, I implore you, O stranger, do not utter such vile heresy, which attacks the very foundation of our freedom! Our supreme law, the principle of Civic Initiative, states that no one can be compelled, constrained, or even coaxed to do what he does not wish. Who, then, would dare expropriate the Eminents' factories, it being their will to enjoy possession of same? That would be the most horrible violation of liberty imaginable. Now, then, to continue, the New Machines produced an abundance of extremely cheap goods and excellent food, but the Drudgelings bought nothing, for they had not the wherewithal . . . "

"But, my dear Phool!" I cried. "Surely you do not claim that the Drudgelings did this voluntarily? Where was your liberty, your civic freedom?!" "Ah, worthy stranger", sighed the Phool, "the laws were still observed, but they say only that the citizen is free to do whatever he wants with his property and money; they do not say where he is to obtain them. No one oppressed the Drudgelings, no one forced them to do anything; they were completely free and could do what they pleased, yet instead of rejoicing at such freedom they died off like flies . . . . . "

. . .